

Polynomial Regression

Machine Learning Assignment

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GitHub Repository: github.com/TheOrangeCarXX/polynomial_regression

1 Introduction

This assignment uses polynomial regression to predict a continuous target y for two personalized datasets, `var1` and `var2`. Each dataset has 1000 training samples with target values and 1000 test samples without target values.

The first problem has six input variables and permits polynomial degrees up to 10. The second has three input variables and permits degrees up to 20. I compared the allowed degrees using cross-validation, selected the model settings, and generated a prediction file for each test dataset. The evaluation metrics are mean squared error (MSE) and the coefficient of determination (R^2).

2 Methodology

For both problems, I used polynomial features, standardization, and Ridge regression. The expansion uses total degree: a term is included when the sum of its input exponents is at most the chosen degree. For example, three inputs at degree two give

$$1, x_1, x_2, x_3, x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2.$$

In the code, the expansion omits the constant column because Ridge fits the intercept separately.

Higher degrees create many related polynomial columns. Ridge adds a penalty on the squared coefficients, controlled by α , to limit large coefficients. The intercept is not penalized. Before regression, `StandardScaler` subtracts each column's training mean and divides by its training standard deviation.

The model-selection procedure was:

1. Split the 1000 labeled rows into 800 development rows and 200 holdout rows, using a shuffled split with random seed 42.
2. Divide the development rows into five cross-validation folds, using random seed 43. Each fold has 640 training rows and 160 validation rows.
3. Evaluate every allowed degree and 50 logarithmically spaced α values from 10^{-4} to 10^4 . Fit the polynomial expansion and scaler separately on each fold's training rows.
4. Select the degree and α with the lowest mean validation MSE across the five folds.
5. Fit those settings on the 800 development rows and evaluate them once on the 200 holdout rows. The holdout does not change the selected settings.
6. Refit the same settings on all 1000 labeled rows and predict the corresponding test dataset.

The search reuses one singular value decomposition per degree and fold to calculate the Ridge solutions for all 50 α values. The final models use a scikit-learn pipeline containing `PolynomialFeatures`, `StandardScaler`, and `Ridge`.

MSE measures the average squared prediction error, so smaller values are better. R^2 compares prediction error with the variation of the target about its mean, so larger values are better. I used MSE to select the settings and recorded R^2 alongside it.

3 Problem 1: var1

3.1 Dataset

The inputs are $x_1, x_2, x_3, x_4, x_5, x_6$. The training file contains these six columns and y ; the test file contains only the inputs. Both files have 1000 rows. All degrees from 1 to 10 were evaluated.

3.2 Degree Selection

For each degree, Table 1 reports the α with the lowest mean validation MSE. The error decreases up to degree 5 and then increases at the higher degrees, even after choosing a suitable α for each degree. Degree 5 gives the lowest MSE in this search.

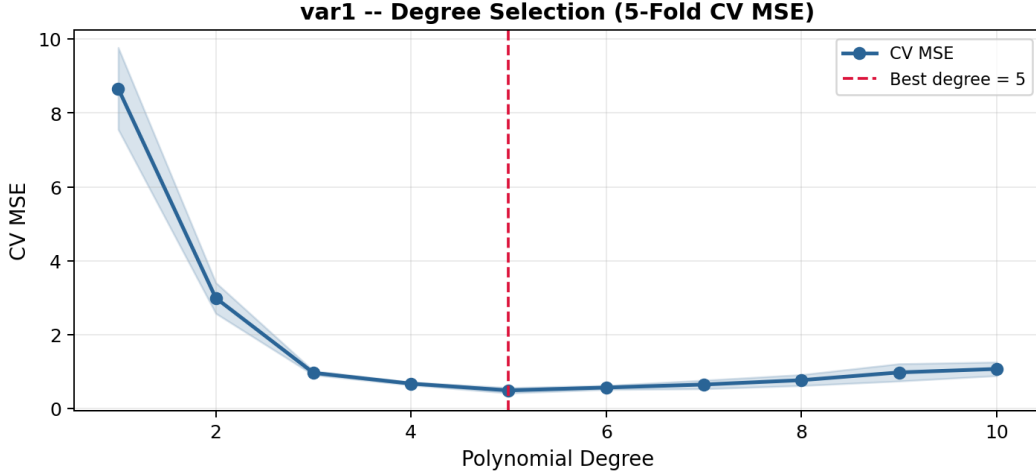


Figure 1: Validation MSE for **var1**. The shaded region shows one standard deviation across the five folds. The MSE axis is logarithmic.

Degree	Selected α	Mean CV MSE	Mean CV R^2
1	51.794747	8.308984	0.093521
2	11.513954	2.786435	0.696228
3	5.428675	1.038719	0.886137
4	11.513954	0.724363	0.919839
5	24.420531	0.528587	0.941952
6	51.794747	0.638766	0.929634
7	109.854114	0.713834	0.921486
8	232.995181	0.847767	0.906211
9	339.322177	0.963495	0.893691
10	494.171336	1.104303	0.877910

Table 1: Five-fold validation results on the 800 development rows for **var1**. Each row uses that degree's best α .

The selected settings are degree 5 and $\alpha = 24.4205309455$. The mean CV MSE is 0.528587, with a fold standard deviation of 0.117995. This standard deviation describes variation between the fold scores; it is not a confidence interval.

3.3 Final Model

The `var1` model uses:

- Polynomial degree: 5.
- Ridge regularization: $\alpha = 24.4205309455$.
- Polynomial columns: 461, plus a separate intercept.

The number of terms including the constant is $\binom{6+5}{5} = 462$.

After selecting the settings, I fitted the model on the 800 development rows and measured its error on the 200 holdout rows. I then fitted a new model on all 1000 labeled rows for test prediction. Table 2 separates these results.

Evaluation	MSE	R^2
Mean five-fold validation	0.528587	0.941952
Development training, 800 rows	0.193723	0.978955
Holdout, 200 rows	0.445705	0.962392
Final model training, 1000 rows	0.203276	0.979125

Table 2: Performance of the selected `var1` configuration. The holdout score belongs to the 800-row model; the final training score belongs to the full-data refit.

3.4 Residual Analysis

A residual is the observed target minus its fitted value. Figure 2 shows residuals for the final model on the full training dataset.

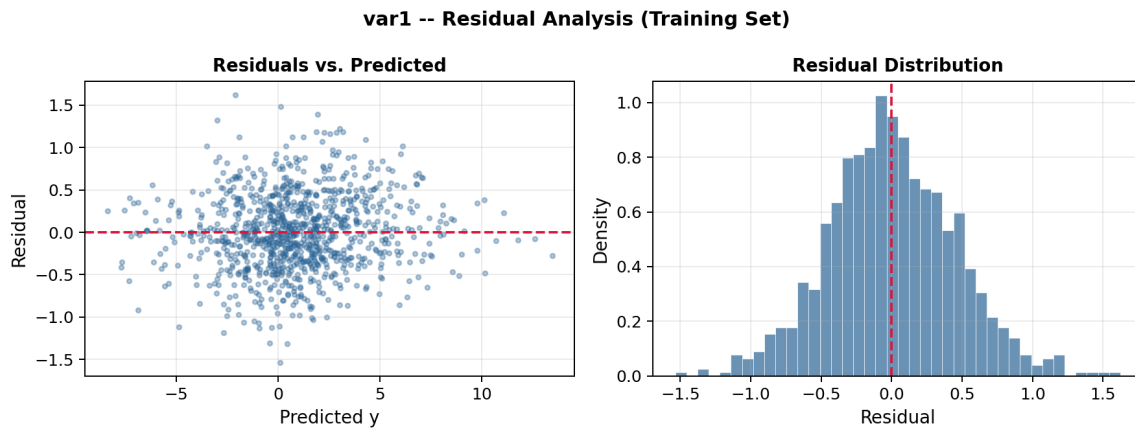


Figure 2: Training residuals for the final degree-5 Ridge model for `var1`.

Most residuals lie near zero, with fewer large positive or negative errors. The residual-versus-fitted plot has no strong curved pattern. These plots describe the training fit. The holdout MSE of 0.445705 is the separate assessment on rows excluded from model selection.

4 Problem 2: var2

4.1 Dataset

The `var2` inputs are x_1, x_2, x_3 . The training file contains 1000 labeled rows, and the test file contains 1000 input rows. All degrees from 1 to 20 were evaluated.

4.2 Degree Selection

The error falls sharply at the lower degrees and changes much less after degree 8. Degree 12 gives the lowest mean validation MSE, although degrees 10 and 14 have close scores. The choice is based on the minimum across the evaluated degree-and- α pairs.

Degree	Selected α	Mean CV MSE	Mean CV R^2
1	16.768329	29.084166	0.256644
2	3.727594	19.929319	0.488921
3	0.568987	9.972992	0.744494
4	0.000100	3.439143	0.910998
5	0.126486	1.432758	0.963236
6	0.001389	0.611912	0.984142
7	0.000100	0.350286	0.991030
8	0.059636	0.281927	0.992704
9	0.268270	0.289924	0.992528
10	0.828643	0.265154	0.993149
11	1.206793	0.273715	0.992942
12	1.757511	0.262967	0.993214
13	2.559548	0.273047	0.992954
14	3.727594	0.270800	0.993015
15	3.727594	0.280955	0.992748
16	5.428675	0.284311	0.992667
17	5.428675	0.293666	0.992421
18	5.428675	0.301023	0.992236
19	5.428675	0.309668	0.992012
20	7.906043	0.319286	0.991766

Table 3: Five-fold validation results on the 800 development rows for `var2`. Each row uses that degree's best α .

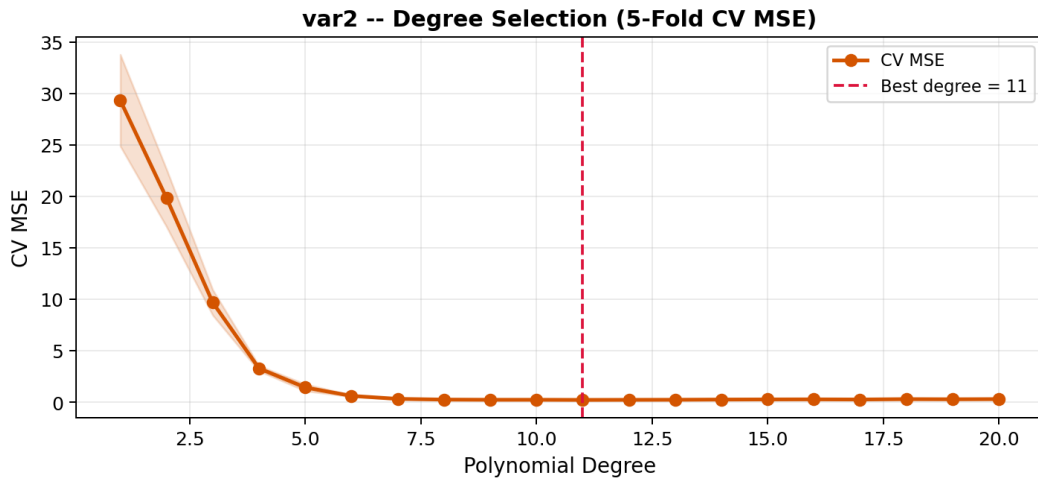


Figure 3: Validation MSE for `var2`. The logarithmic axis keeps the higher-degree results visible. The shaded region shows one standard deviation across the five folds.

4.3 Final Model

The **var2** model uses:

- Polynomial degree: 12.
- Ridge regularization: $\alpha = 1.7575106249$.
- Polynomial columns: 454, plus a separate intercept.

The degree-12 expansion contains $\binom{3+12}{12} = 455$ terms including the constant. The selected settings were assessed on the holdout and then refitted on the full training dataset, using the same procedure as **var1**.

Evaluation	MSE	R^2
Mean five-fold validation	0.262967	0.993214
Development training, 800 rows	0.157365	0.995993
Holdout, 200 rows	0.254243	0.994152
Final model training, 1000 rows	0.160039	0.996019

Table 4: Performance of the selected **var2** configuration. The holdout uses the 800-row model, before the full-data refit.

The holdout MSE is close to the mean validation MSE for this split. Degree 12 is the lowest-error choice in the search, but the small differences among nearby degrees do not establish a large performance advantage.

4.4 Residual Analysis

Figure 4 shows the final model’s residuals on the 1000 training rows.

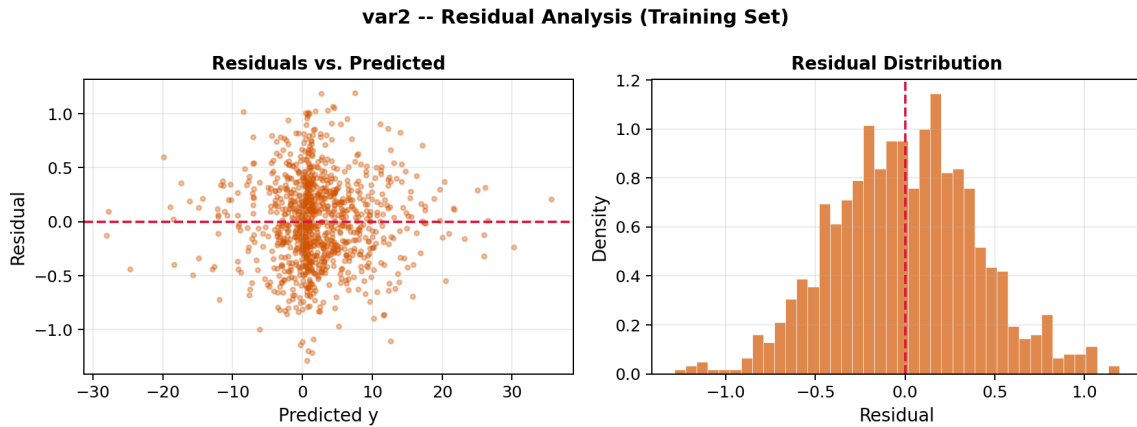


Figure 4: Training residuals for the final degree-12 Ridge model for **var2**.

The histogram is centered near zero, and the scatter plot does not show a strong curved trend. A few rows have larger residuals on either side. The plots are useful for checking the fitted model, while the holdout gives the separate error measurement.

5 Inference and Prediction

The final pipelines use all 1000 labeled rows. For each test row, the pipeline applies the chosen polynomial expansion, scales the columns using the final training statistics, and applies the fitted Ridge model:

$$X_{\text{test}} \longrightarrow \phi_d(X_{\text{test}}) \longrightarrow Z_{\text{test}} \longrightarrow \hat{y}_{\text{test}}.$$

Here, ϕ_d denotes the degree- d polynomial expansion, and Z_{test} denotes its standardized columns. The scaler is not fitted again on the test data.

The output files are:

- BT2024103_pred_var1.csv
- BT2024103_pred_var2.csv

Each file has a single y column and 1000 rows, with no index column. Predictions preserve the order of the supplied test rows. Since the test targets are hidden, test MSE and R^2 cannot be calculated locally.

6 Implementation

The training and inference code is in `ml_assignment_solution.py`. It loads and checks the four CSV files, performs model selection, evaluates the holdout, fits the final pipelines, and writes the prediction files. It also creates the four figures used in this report.

The saved pipelines are in `models/`. The complete search results, best settings for each degree, evaluation metrics, and split seeds are in `results/`. Input checks require the expected columns, 1000 rows per file, and finite numeric values.

To repeat training and prediction, place the four input CSVs together in `data/` and run:

```
python -m pip install -r requirements.txt
python ml_assignment_solution.py --data-dir data
```

To use the saved final models without repeating model selection, run:

```
python ml_assignment_solution.py --data-dir data --predict-only
```

The input folder can also be supplied as another path through `--data-dir`. The code therefore does not depend on a personal Windows directory.

7 Conclusion

The search selected degree 5 with $\alpha = 24.4205309455$ for `var1`, and degree 12 with $\alpha = 1.7575106249$ for `var2`. Their holdout MSE values were 0.445705 and 0.254243, respectively.

These are the lowest-CV-MSE settings among the configurations evaluated. The holdout was kept separate from selection, although its result still depends on the particular 200-row split. Both selected models were refitted on the complete labeled datasets and used to produce the required prediction CSV files.